General Software engineering Role Case study

This is a generic case study for the general software engineering positions for all teams. You will be selected based on your ability to follow specifications, work on a component that fits into a larger machine, attention to detail, communication, and initiative. You will be considered even if you submit a partially complete study. If you are submitting a partial work, please state it is incomplete and provide a reason.

Background

In order to navigate and deliver packages within a warehouse, autonomous robots operating within the warehouse require careful closed loop control in order to effectively detect and react to external stimuli within such an unpredictable environment. Warehouse robots should act effectively to ensure timely deliveries around the warehouse while being equipped with proper safety protocol to prevent injury.

Your task

Your goal is to design a control system for an autonomous robot that is able to navigate a warehouse environment and effectively collect and deliver packages. Your robot will come included with:

- IR sensor to detect robot path marked on warehouse floor (0 dark - 1027 light)
- Ultrasonic Sensor to detect obstacles (time taken for pulse to return μs)
- Encoders on motors to detect the number of revolutions.
- ESP32 Microcontroller as the brain

Your objective is to design the navigation algorithm of the robot so it can travel from one designated package zone within the warehouse to the other following a designated path marked on the floor. This robot must be able to detect and avoid any obstacles in its path. Your algorithm should include all your behavioral logic (eg. finite state machine or path planning) and lower level logic (eg. kinematics to convert velocity into wheel speeds in each motor or filter for noisy signals).

You do not need to design the hardware abstraction layer and can assume one is already provided to convert your outputs into machine inputs and to convert sensor data into numerical outputs.

Considerations

- Is your robot able to effectively stop at obstacles to prevent damage or injury?
- Is your robot able to effectively stay on course or readjust if it starts moving off course?

- Does your robot include failsafes to prevent errors such as sending motor speeds that are impossibly fast or trying to complete two tasks at the same time?
- Is your code consistent and readable?
- Is your project hierarchy easy to follow?

Your submission

- Your project should be submitted as a zip file with your name and student ID.
- Files should be clearly labelled in your project.
- Your project should be written in C++ or C
- A short PDF document containing your brief rationale of your design, under 300 words. Show us why you made the decisions you made, what your potential material choices are, where you struggled (especially if your design is incomplete), etc.